

Team-1
5-members

AI Friday Season 2 – Regional Finale

Manufacturing Quality Control Defect Detection Agent

PROBLEM STATEMENT

Manufacturing plants require efficient quality control processes to detect defects early and reduce rework or scrap costs. Manual inspections are labor-intensive and subject to human error, causing inconsistent quality outcomes. Existing automation solutions are often limited to specific defect types and lack integration with production data and feedback loops. An AI multi-agent system that analyzes sensor data and inspection images, integrates with MES systems, and provides real-time defect detection and alerts can significantly enhance quality control effectiveness.

Data Considerations:

Sensor readings (temperature, vibration, etc.) and high-resolution inspection images; production metadata from MES; historical defect records; moderate to high data volume with real-time streaming; preprocessing includes noise filtering and image normalization; privacy for proprietary production data; synthetic defect datasets with labeled anomalies for development.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

A multi-agent AI system featuring agents for sensor/image data ingestion, defect detection using computer vision and pattern recognition, and alert management. Features include a quality control dashboard with real-time defect visualization, trend analysis, and reporting. Integrations with MES APIs enable automated data flow and incident logging. Success metrics focus on detection accuracy, inspection time reduction, and defect rate improvement. Demonstrations include live defect detection scenarios, multi-agent workflow, and comprehensive documentation with test suites.

Feel free to explore creative solutions that align with the problem context and objectives.